

OCTOBER 2026

Predictive Mock Test SOLUTION BOOK

CHEMISTRY + BIOLOGY PATHWAY

81

questions

3

modules

40

minutes per module

MODULES IN THIS BOOK

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03	Biology	58

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MODULE 01

Mathematics 1

QUESTIONS & WORKED SOLUTIONS

Chemistry + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A rectangular card has a length recorded as 12 cm and a width recorded as 8 cm. Both measurements are rounded to the nearest centimetre. What is the least upper bound for the ratio of its actual length to its actual width?

A $\frac{3}{2}$

B $\frac{5}{3}$

C $\frac{23}{15}$

D $\frac{25}{17}$

E $\frac{25}{16}$

Worked solution

Answer B

Write the actual length and width as L and W . Their bounds are $11.5 \leq L < 12.5$ and $7.5 \leq W < 8.5$. The largest ratios use a length approaching 12.5 and the smallest width, 7.5. Thus $\frac{L}{W} < \frac{12.5}{7.5} = \frac{5}{3}$, and ratios can approach $\frac{5}{3}$ as closely as desired. The least upper bound is $\frac{5}{3}$.

Trap: using both upper bounds. To make a positive ratio as large as possible, make its numerator large and its denominator small. The upper bound need not be attained.

Question 02

For a real number k , the equation $x^2 - (k + 4)x + 2k = 0$ has one root strictly between 0 and 1, and its other root is greater than 2. What is the set of all possible values of k ?

A $k < 0$

B $0 \leq k \leq 3$

C $k > 3$

D $0 < k < 3$

E $k < 0$ or $k > 3$

Worked solution

Answer D

Let $p(x) = x^2 - (k + 4)x + 2k$. A root in $(0, 1)$ requires $p(0) > 0$ and $p(1) < 0$, giving $k > 0$ and $k < 3$. Also $p(2) = -4 < 0$, so after the first crossing the other root is beyond 2.

Trap: checking only the discriminant and forgetting where the roots lie.

Question 03

What is the value of x that satisfies $\frac{9^{x+1} 27^{2-x}}{3^{x-1}} = 81$?

A $\frac{5}{2}$

B $\frac{3}{2}$

C 2

D $\frac{1}{2}$

E $\frac{7}{2}$

Worked solution

Answer A

Write every number as a power of 3. The equation becomes $3^{2(x+1)+3(2-x)-(x-1)} = 3^4$.
Hence $9 - 2x = 4$, giving $x = \frac{5}{2}$.

Trap: subtract the whole denominator exponent $x - 1$, so its contribution is $-x + 1$.

Question 04

The least positive integer n leaves remainder 2 when divided by 5, and remainder 3 when divided by 7. What is the remainder when $n^2 + 4n + 1$ is divided by 35?

A 3

B 5

C 18

D 13

E 8

Worked solution

Answer E

The candidates leaving remainder 2 on division by 5 are 2, 7, 12, 17, The first that leaves remainder 3 on division by 7 is $n = 17$. Then $n^2 + 4n + 1 = 358 = 10 \cdot 35 + 8$, so the remainder is 8.

Question 05

A container holds 4 red, 3 blue and 2 green counters. Three counters are drawn without replacement. Given that exactly one of the three is green, what is the probability that the first counter drawn is red?

A $\frac{1}{7}$

B $\frac{8}{21}$

C $\frac{2}{9}$

D $\frac{4}{9}$

E $\frac{1}{2}$

Worked solution

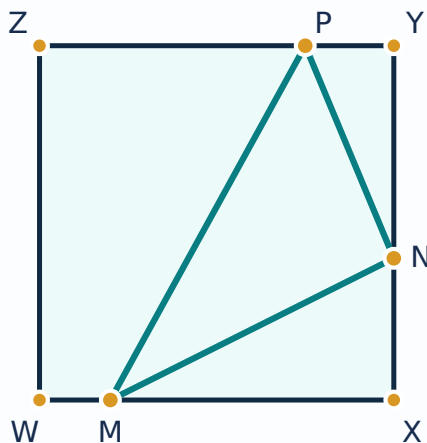
Answer B

Given that exactly one of the three counters is green, the green counter is equally likely to occupy any of the three positions. Therefore the first counter is non-green with probability $\frac{2}{3}$. Conditional on its being non-green, each of the seven non-green counters is equally likely to be first, and four are red. Hence the required probability is $\frac{2}{3} \times \frac{4}{7} = \frac{8}{21}$.

Trap: conditioning changes the sample space; the unconditional probability $\frac{4}{9}$ is not the answer.

Question 06

$WXYZ$ is a square of side 1. Points M, N, P lie on WX, XY, YZ , respectively, with $WM : MX = 1 : 4$, $XN : NY = 2 : 3$, and $YP : PZ = 1 : 3$. What is the area of triangle MNP ?



Schematic diagram; not to scale.

A $\frac{1}{5}$

B $\frac{1}{4}$

C $\frac{29}{100}$

D $\frac{3}{10}$

E $\frac{7}{20}$

Worked solution

Answer C

The ratios give $WM = \frac{1}{5}$, $MX = \frac{4}{5}$, $XN = \frac{2}{5}$, $NY = \frac{3}{5}$, $YP = \frac{1}{4}$, and $ZP = \frac{3}{4}$. Subtract the two right triangles MXN , NYP , and the trapezium $WMPZ$ from the unit square.

The two triangle areas are $\frac{1}{2} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{4}{25}$ and $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{40}$. The trapezium has parallel sides WM and ZP , separated by distance 1, so its area is $\frac{1}{2}(\frac{1}{5} + \frac{3}{4}) = \frac{19}{40}$. Thus the required area is $1 - \frac{4}{25} - \frac{3}{40} - \frac{19}{40} = \frac{29}{100}$.

Tip: subtract familiar shapes from the square; no general coordinate-area formula is needed.

Question 07

Pumps A and B fill a tank at constant rates. In a trial, increasing A's rate by 20% and decreasing B's rate by 10% leaves their combined rate unchanged.

At its original rate, A alone fills the tank in 30 minutes. The empty tank is now filled using both pumps at their original rates. After 4 minutes, B is switched off and A completes the filling.

What is the total time taken to fill the tank?

A 14 minutes

B 18 minutes

C 20 minutes

D 22 minutes

E 24 minutes

Worked solution

Answer D

Let the original filling rates be a and b tanks per minute. The trial gives

$$1.2a + 0.9b = a + b \Rightarrow 0.2a = 0.1b \Rightarrow b = 2a.$$

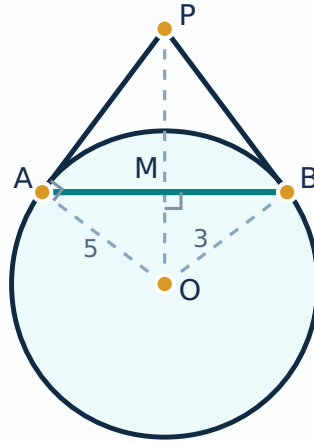
Since $a = 1/30$, the combined rate is $3a = 1/10$ tank per minute. In the first 4 minutes, the pumps fill $4/10 = 2/5$ of the tank.

A fills the remaining $3/5$ in $(3/5) \times 30 = 18$ minutes. The total is $4 + 18 = 22$ minutes.

Useful approach: Use the unchanged combined rate to find the ratio of the two original rates before calculating any times.

Question 08

A circle has centre O and radius 5. Chord AB is 3 units from O , measured perpendicularly. The tangents at A and B meet at P . What is OP ?



Schematic diagram; M is the midpoint of AB .

A $\frac{25}{3}$

B $\frac{25}{6}$

C 5

D 10

E 15

Worked solution

Answer A

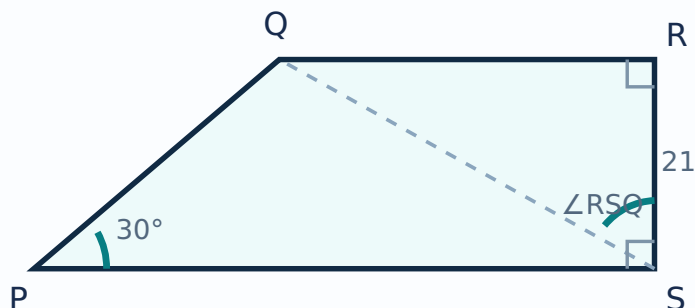
Let M be the midpoint of chord AB . The perpendicular from O to AB passes through M , and O, M, P are collinear by symmetry. We have $OM = 3$ and $OA = 5$.

Triangle OMA is right-angled at M , while triangle OAP is right-angled at A , because a radius is perpendicular to a tangent. They share the angle at O , so they are similar.

Hence $\frac{OM}{OA} = \frac{OA}{OP}$, giving $\frac{3}{5} = \frac{5}{OP}$ and $OP = \frac{25}{3}$.

Question 09

$PQRS$ is a right trapezium with $PS \parallel QR$, right angles at R, S , $RS = 21$, $\angle QPS = 30^\circ$, and $\tan \angle RSQ = \frac{4}{3}$. What is PS ?



Schematic diagram; diagonal SQ is shown for the given angle.

A $28 + 21\sqrt{3}$

B $21 + 21\sqrt{3}$

C $21 + 28\sqrt{3}$

D $49 + \sqrt{3}$

E $28 + 7\sqrt{3}$

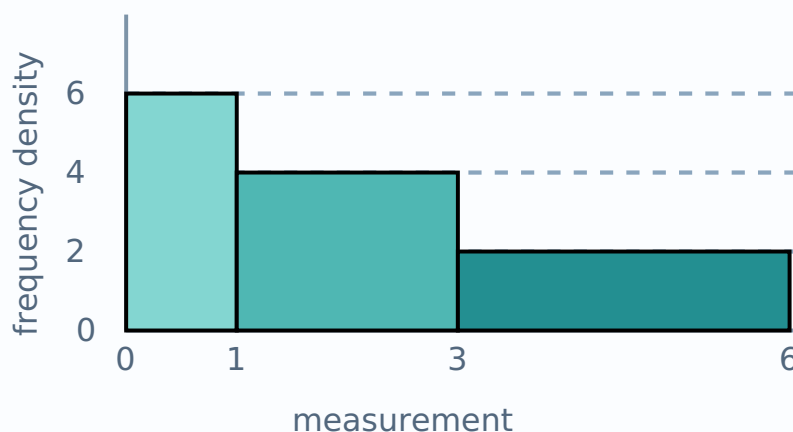
Worked solution

Answer A

In right triangle QRS , $QR/RS = 4/3$, so $QR = 28$. The horizontal distance from P to the vertical through Q is $21/\tan 30^\circ = 21\sqrt{3}$. Therefore $PS = 28 + 21\sqrt{3}$.

Question 10

A histogram represents exactly 20 measurements. Its class intervals are $0 \leq x < 1$, $1 \leq x < 3$, and $3 \leq x \leq 6$, where x is the measurement. The frequency densities are shown below.



The unequal widths are part of the histogram.

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

Worked solution

Answer B

Frequency equals class width multiplied by frequency density, so the three frequencies are $1 \times 6 = 6$, $2 \times 4 = 8$, and $3 \times 2 = 6$. The total of all measurements is $20(1.8) = 36$.

Only the six observations in $[3, 6]$ can be at least 5. If all six were, the total would be at least $6(5) + 8(1) + 6(0) = 38$, which is impossible. Five are achievable: take five values 5, one value 3, eight values 1, and six values 0. These respect the stated class boundaries and sum to 36. The maximum number is therefore 5.

Question 11

Two fair six-sided dice are rolled. Given that at least one die shows 4, what is the probability that the sum of the scores is at least 9?

A $\frac{1}{3}$

B $\frac{2}{5}$

C $\frac{4}{11}$

D $\frac{5}{12}$

E $\frac{1}{2}$

Worked solution

Answer C

Given that at least one die shows 4, there are $6 + 6 - 1 = 11$ equally likely ordered outcomes. The outcomes with sum at least 9 are (4, 5), (4, 6), (5, 4), (6, 4). Therefore the probability is $4/11$.

Question 12

For positive integers n , define $a_n = n^2 + 3n + 2$. How many of the first 120 terms are divisible by 12?

A 32

B 36

C 44

D 40

E 48

Worked solution

Answer D

Factor $a_n = (n+1)(n+2)$. Divisibility by 12 requires divisibility by both 3 and 4.

For the factor 4, the even member of the two consecutive integers must be a multiple of 4. Thus n leaves remainder 2 or 3 when divided by 4.

For the factor 3, n must leave remainder 1 or 2 when divided by 3. In each block of 12 consecutive values of n , both requirements hold at remainders 2, 7, 10, 11.

There are $120/12 = 10$ complete blocks, each with 4 successes, giving 40 terms.

Question 13

The graph of $y = g(x)$ is made from straight-line segments joining, in order, the points $(-4, 3)$, $(-1, -3)$, $(2, 1)$, $(5, -2)$, $(7, 2)$. What is the sum of the x -coordinates of all its intersections with the x -axis?

A $\frac{29}{4}$

B $\frac{31}{4}$

C 8

D $\frac{33}{4}$

E $\frac{35}{4}$

Worked solution

Answer B

On the first segment, the heights 3 and -3 put the crossing halfway between $x = -4$ and $x = -1$, at $-\frac{5}{2}$. On the second segment, the height rises from -3 to 1, so zero is reached three quarters of the way from -1 to 2: $-1 + \frac{3}{4}(3) = \frac{5}{4}$.

On the third segment, the height falls from 1 to -2 , so the crossing is one third of the way from 2 to 5, at 3. The final segment crosses halfway from 5 to 7, at 6. Their sum is $-\frac{5}{2} + \frac{5}{4} + 3 + 6 = \frac{31}{4}$.

Question 14

For a real parameter a , consider $x^2 - 2(a+1)x + a^2 - 1 = 0$. For which values of a does this equation have one positive root and one negative root?

A $a \leq -1$

B $-1 \leq a < 0$

C $-1 < a < 1$

D $0 < a \leq 1$

E $a < -1$ or $a > 1$

Worked solution

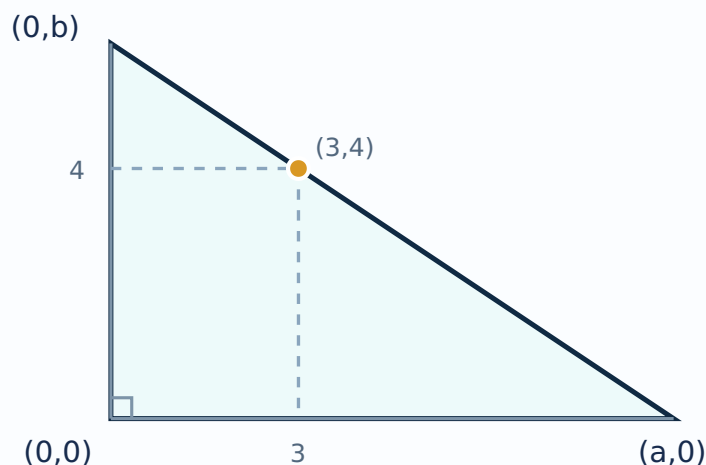
Answer C

Writing a monic quadratic as $(x-r)(x-s)$ shows that its constant term is the product rs . Here the product is $a^2 - 1$. Opposite signs require a negative product, so $a^2 - 1 < 0$, giving $-1 < a < 1$.

This condition is also sufficient: the upward-opening quadratic is negative at $x = 0$ and positive for sufficiently large positive or negative x , so it crosses the axis once on each side of zero. The endpoints $a = \pm 1$ are excluded because one root is then zero.

Question 15

A right triangle has vertices $(0,0)$, $(a,0)$, $(0,b)$, where $a, b > 0$. The point $(3,4)$ lies on its hypotenuse. What is the minimum possible area of the triangle?



Schematic diagram; not to scale.

A 18

B 20

C 32

D 28

E 24

Worked solution

Answer E

The line through $(a,0)$ and $(0,b)$ has equation $y = b - \frac{b}{a}x$, equivalently $\frac{x}{a} + \frac{y}{b} = 1$. Since $(3,4)$ lies on it, $\frac{3}{a} + \frac{4}{b} = 1$. Set $u = \frac{3}{a}$ and $v = \frac{4}{b}$, so $u + v = 1$ and $ab = \frac{12}{uv}$.

Because $(u - v)^2 \geq 0$, we have $4uv \leq (u + v)^2 = 1$. Therefore $ab \geq 48$, and the area is at least $\frac{1}{2}(48) = 24$. Equality occurs at $u = v = \frac{1}{2}$, giving $a = 6$ and $b = 8$, so this minimum is attained.

Question 16

A sequence is defined by $a_1 = 1$ and $a_{n+1} = 3a_n + 2$. What is the remainder when a_{2026} is divided by 7?

A 4

B 2

C 1

D 5

E 6

Worked solution

Answer A

The successive remainders on division by 7 are 1, 5, 3, 4, 0, 2, then repeat. Since $2026 = 6 \cdot 337 + 4$, the required remainder is the fourth one in the cycle, namely 4.

Question 17

Points $A = (-4, 7)$, $B = (5, -2)$ and $C = (3, 6)$ are given. Point P lies on AB such that $AP : PB = 2 : 1$. Point Q is the reflection of C in P , meaning a rotation of 180° about P . What is $2x_Q - y_Q$?

A -6**B** -2**C** 2**D** 6**E** 10

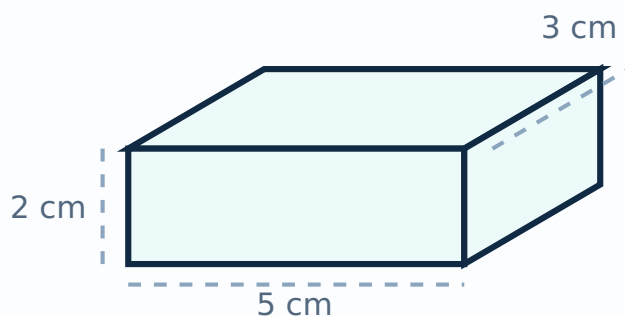
Worked solution

Answer D

$P = A + \frac{2}{3}(B - A) = (-4, 7) + \frac{2}{3}(9, -9) = (2, 1)$. Since P is the midpoint of CQ , $Q = 2P - C = (1, -4)$. Thus $2x_Q - y_Q = 2 + 4 = 6$.

Question 18

A rectangular metal block measures $2\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$ and has mass 270 g . What is its density in kg m^{-3} ?



Schematic block; dimensions are not drawn to scale.

A 90

B 900

C 9000

D 27000

E 90000

Worked solution

Answer C

The block has volume $2 \times 3 \times 5 = 30\text{ cm}^3 = 3 \times 10^{-5}\text{ m}^3$. Its mass is $270\text{ g} = 0.270\text{ kg}$.
Therefore its density is $\frac{0.270}{3 \times 10^{-5}} = 9000\text{ kg m}^{-3}$.

Question 19

Let O, A, B, C be the vertices of a parallelogram, with O as the origin and $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ as adjacent sides. Thus C is the fourth vertex. Point P lies on AC with $AP : PC = 1 : 2$, and Q is the midpoint of OB . Which expression is $\overrightarrow{PQ}$?

A $\mathbf{a} + \frac{1}{6}\mathbf{b}$

B $-\mathbf{a} + \frac{1}{6}\mathbf{b}$

C $-\mathbf{a} - \frac{1}{6}\mathbf{b}$

D $\mathbf{a} - \frac{1}{6}\mathbf{b}$

E $-\mathbf{a} + \frac{1}{3}\mathbf{b}$

Worked solution

Answer B

Because OA and OB are adjacent sides, $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$, so $P = \mathbf{a} + \frac{1}{3}\mathbf{b}$. Also $Q = \frac{1}{2}\mathbf{b}$.
Therefore $\overrightarrow{PQ} = Q - P = -\mathbf{a} + \frac{1}{6}\mathbf{b}$.

Question 20

The line $y = 2x - 1$ is tangent to the graph $y = x^2 - 2kx + k + 3$, where k is real. What is the sum of the two possible values of k ?

A -3

B -2

C 3

D 1

E -1

Worked solution

Answer E

At an intersection, $x^2 - 2(k+1)x + k + 4 = 0$. Completing the square gives $(x - k - 1)^2 = k^2 + k - 3$. A tangent has exactly one point of contact, so the right-hand side must be zero.

Thus $k^2 + k - 3 = 0$, giving $k = \frac{-1+\sqrt{13}}{2}$ or $k = \frac{-1-\sqrt{13}}{2}$. The sum is -1 .

Tip: a repeated root can be recognised by completing the square; differentiation is not required.

Question 21

A quantity P is directly proportional to x^2 and inversely proportional to y . If x is increased by 20% and y is decreased by 20%, by what factor does P change?

A $\frac{4}{5}$

B $\frac{6}{5}$

C $\frac{9}{5}$

D 2

E $\frac{12}{5}$

Worked solution

Answer C

Write $P = c\frac{x^2}{y}$ for some constant c . After the changes, x is multiplied by $6/5$ and y by $4/5$, so P is multiplied by $\frac{(6/5)^2}{4/5} = \frac{9}{5}$.

Question 22

Six cards are numbered 1, 2, 3, 4, 5, 6. Two different cards are selected, with every pair equally likely.

Given that the product of their numbers is even, what is the probability that their sum is greater than 7?

A $\frac{1}{3}$

B $\frac{1}{5}$

C $\frac{5}{12}$

D $\frac{1}{2}$

E $\frac{3}{5}$

Worked solution

Answer C

There are $6 \times 5/2 = 15$ unordered pairs. The product is odd only when both cards are odd. The three odd cards form 3 pairs, so $15 - 3 = 12$ pairs satisfy the condition.

The eligible pairs whose sum exceeds 7 are

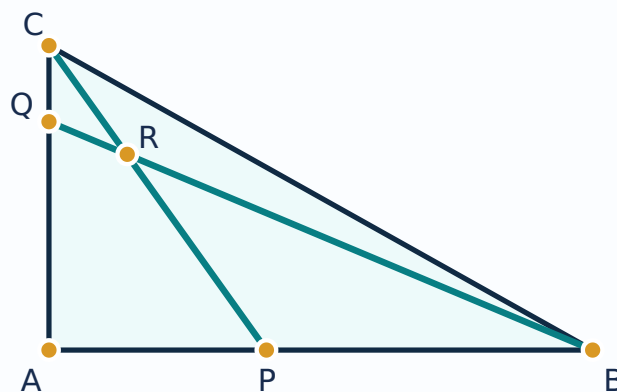
$$(2, 6), (3, 6), (4, 5), (4, 6), (5, 6).$$

The pair (3, 5) has a large enough sum but has an odd product. The required probability is $5/12$.

Useful approach: Apply the given condition to both the denominator and the favourable outcomes.

Question 23

In triangle ABC , let $A = (0, 0)$, $B = (10, 0)$, and $C = (0, 8)$. Point P lies on AB with $AP : PB = 2 : 3$, and point Q lies on AC with $AQ : QC = 3 : 1$. Lines CP and BQ meet at R . What is the x -coordinate of R ?



Schematic diagram; not to scale.

A 1

B $\frac{4}{3}$

C $\frac{5}{3}$

D $\frac{10}{7}$

E 2

Worked solution

Answer D

$P = (4, 0)$, $Q = (0, 6)$. The two lines are $CP : y = 8 - 2x$ and $BQ : y = 6 - \frac{3}{5}x$. Equating gives $2 = \frac{7}{5}x$, hence $x = 10/7$.

Question 24

Box A is chosen with probability $\frac{2}{3}$, and box B with probability $\frac{1}{3}$. Box A contains 2 red and 3 blue counters; box B contains 4 red and 1 blue counter. Two counters are drawn without replacement from the chosen box. Given that the two counters have different colours, what is the probability that box A was chosen?

A $\frac{1}{2}$

B $\frac{2}{3}$

C $\frac{7}{10}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer D

Let D mean that the two drawn counters have different colours. From box A , the red-blue and blue-red branches give $P(D | A) = 2(\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. For B , $P(D | B) = 2(\frac{4}{5})(\frac{1}{4}) = \frac{2}{5}$.

The probability of choosing A and then drawing different colours is $\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$. The probability of different colours overall is $\frac{2}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{8}{15}$. Of these different-colour outcomes, the fraction originating from A is $\frac{2/5}{8/15} = \frac{3}{4}$.

Question 25

How many ordered pairs of positive integers (a, b) satisfy $ab = 360$, have highest common factor 1, and have $a + b$ a multiple of 7?

A 2

B 0

C 6

D 8

E 4

Worked solution

Answer E

Since $360 = 2^3 \cdot 3^2 \cdot 5$ and a, b are coprime, the prime powers 8, 9, 5 must be assigned wholly to one factor. The possible unordered splits are $1 + 360, 8 + 45, 9 + 40, 72 + 5$; only $9 + 40 = 49$ and $72 + 5 = 77$ are multiples of 7. Both orders count, giving 4.

Question 26

Data set A has mean 14, and data set B has mean 20. The mean of all observations in the two data sets combined is 16. An observation of value 10 from A is exchanged with an observation of value 34 from B . The two data sets then have the same mean. How many observations are there altogether?

A 6

B 12

C 18

D 24

E 36

Worked solution

Answer C

Exchanging the two observations changes neither the combined total nor the size of either data set. The combined mean remains 16, so the two equal new means must both be 16.

The total in A increases by $34 - 10 = 24$, while its mean increases by $16 - 14 = 2$. Therefore A contains $24/2 = 12$ observations. The total in B decreases by 24, while its mean decreases by $20 - 16 = 4$, so B contains $24/4 = 6$ observations. Altogether there are $12 + 6 = 18$.

Trap: averaging the two original means without accounting for the different data set sizes.

Question 27

Two geometrically similar solid objects are made from the same uniform material. Their total surface areas are in the ratio $4 : 9$, smaller to larger. The mass of the larger object is 95 g greater than the mass of the smaller object. What is the mass of the smaller object?

A 40 g

B 20 g

C 76 g

D 95 g

E 190 g

Worked solution

Answer A

The surface area ratio $4 : 9$ gives a length ratio $2 : 3$. The volume ratio is therefore $2^3 : 3^3 = 8 : 27$. Because the objects are made from the same uniform material, their masses also have ratio $8 : 27$.

The difference of $27 - 8 = 19$ ratio parts is 95 g, so one part is 5 g. The smaller mass is $8 \times 5 = 40$ g.

Trap: applying the surface area ratio directly to mass gives 76 g; using the length ratio gives 190 g.

MODULE 02

Chemistry

QUESTIONS & WORKED SOLUTIONS

Chemistry + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A dilute solution of hydrochloric acid dissociates completely. Two separate portions are diluted with water. The observations are shown below.

Dilution	Methyl orange	Bromothymol blue
Final volume is 10 times the original volume	red	yellow
Final volume is 100 times the original volume	yellow	yellow

Assume sharp indicator changes: methyl orange is red below pH 3.7 and yellow above pH 3.7; bromothymol blue is yellow below pH 6.5 and blue above pH 6.5. None of the solutions is exactly at a transition pH. Ignore the contribution of water to the hydrogen-ion concentration. For these solutions, $[H^+] = 10^{-n} \text{ mol dm}^{-3}$ corresponds to pH n .

Which could be the original hydrogen-ion concentration, in mol dm^{-3} ?

A 10^{-1}

B 10^{-2}

C 10^{-3}

D 10^{-4}

E 10^{-5}

Worked solution

Answer B

A tenfold dilution lowers the hydrogen-ion concentration by a factor of 10, so it raises pH by 1. A hundredfold dilution raises pH by 2.

If the original pH is p , the methyl-orange observations give $p + 1 < 3.7$ and $p + 2 > 3.7$. Therefore $1.7 < p < 2.7$.

Of the options, only $10^{-2} \text{ mol dm}^{-3}$ has a pH in this interval: pH 2. Its diluted solutions have pH 3 and 4, both below the bromothymol-blue transition. **Answer: B.**

Question 02

Atoms of elements X and Y have electron configurations 2,8,2 and 2,8,7 respectively. Z is the element immediately below Y in the same group of the Periodic Table.

Which row gives the formula of the ionic compound formed by X and Y, and the expected result when Y_2 is added to an aqueous solution of NaZ?

A XY; Y_2 displaces Z from Z^- ions.

B XY_2 ; no displacement occurs.

C X_2Y ; Y_2 displaces Z from Z^- ions.

D XY_2 ; Y_2 displaces Z from Z^- ions.

E X_2Y ; no displacement occurs.

Worked solution

Answer D

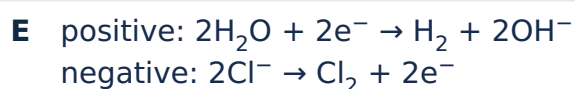
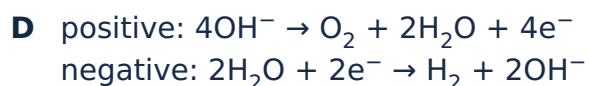
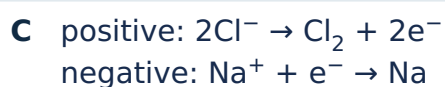
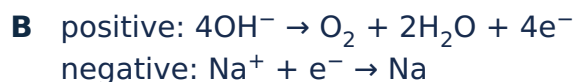
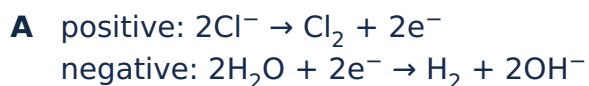
X loses its two outer electrons to form X^{2+} . Y gains one electron to form Y^- . Two Y^- ions balance each X^{2+} , so the formula is XY_2 .

Y is a Group 17 halogen. Halogen reactivity decreases down the group, so Y is more reactive than Z. The displacement is $Y_2 + 2Z^- \rightarrow 2Y^- + Z_2$. **Answer: D.**

Question 03

Concentrated aqueous sodium chloride is electrolysed using inert electrodes. The gas at the positive electrode bleaches damp litmus paper. The gas at the negative electrode gives a squeaky pop with a lighted splint.

Which row is correct?



Worked solution

Answer A

The positive electrode is the anode. Chloride ions lose electrons there, producing the chlorine identified by its bleaching action: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$.

The negative electrode is the cathode. In aqueous sodium chloride, water is reduced to hydrogen rather than sodium ions being reduced to sodium: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$. Hydrogen gives the squeaky pop. **Answer: A.**

Question 04

Two pure substances have the following properties.

Property	P	Q
Melting point / °C	-114	801
Boiling point / °C	-85	1465
Conducts when pure and liquid?	no	yes
Conducts when dissolved in water?	yes	yes

Which explanation is consistent with these observations?

- A** P and Q both consist of ions; the ions in pure liquid P cannot move.
- B** P contains delocalised electrons only in its gas phase; Q contains molecules.
- C** P consists of small molecules that form ions on reacting with water; Q contains ions that become mobile when molten or dissolved.
- D** P has a giant covalent structure that breaks into ions on melting; Q conducts through delocalised electrons.
- E** P consists of small molecules whose electrons become delocalised in water; Q consists of small molecules that become mobile on melting.

Worked solution

Answer C

P has very low melting and boiling points and does not conduct as a pure liquid, consistent with small covalent molecules. Its aqueous solution can conduct because those molecules react with water to form ions; aqueous hydrogen chloride is an example.

Q has a high melting point and conducts when molten or dissolved, as expected for an ionic substance with mobile ions. Conductivity of a solution alone does not show that the undissolved substance was ionic. **Answer: C.**

Question 05

A compound contains only carbon, hydrogen and oxygen. Complete combustion of 0.900 g of the compound produces 1.32 g of carbon dioxide and 0.540 g of water. Its relative molecular mass is 90.

Use $A_r(\text{C}) = 12$, $A_r(\text{H}) = 1$ and $A_r(\text{O}) = 16$. What is its molecular formula?

A CH_2O

B $\text{C}_2\text{H}_4\text{O}_2$

C $\text{C}_3\text{H}_8\text{O}_2$

D $\text{C}_6\text{H}_{12}\text{O}_6$

E $\text{C}_3\text{H}_6\text{O}_3$

Worked solution

Answer E

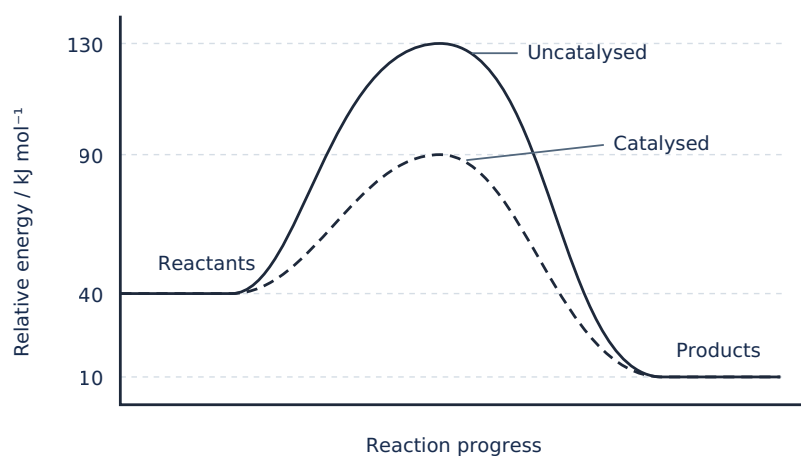
Carbon dioxide gives $n(\text{C}) = 1.32/44 = 0.0300$ mol, so the sample contains 0.360 g of carbon. Water gives $n(\text{H}) = 2 \times 0.540/18 = 0.0600$ mol, equivalent to 0.0600 g of hydrogen.

The oxygen in the original sample therefore has mass $0.900 - 0.360 - 0.0600 = 0.480$ g, or 0.0300 mol. The atom ratio C:H:O is $0.0300:0.0600:0.0300 = 1:2:1$.

The empirical formula CH_2O has formula mass 30. Since $90/30 = 3$, the molecular formula is $\text{C}_3\text{H}_6\text{O}_3$. **Answer: E.**

Question 06

The diagram shows two pathways between the same reactants and products.



Which row gives the activation energy of the **catalysed reverse reaction** and the energy change, ΔH , of the **forward reaction**, in kJ mol^{-1} ?

A Activation energy 50; $\Delta H = -30$

B Activation energy 80; $\Delta H = -30$

C Activation energy 80; $\Delta H = +30$

D Activation energy 120; $\Delta H = -30$

E Activation energy 90; $\Delta H = +30$

Worked solution

Answer B

The reverse reaction starts at the product level, 10 kJ mol^{-1} . The catalysed pathway reaches 90 kJ mol^{-1} , so its reverse activation energy is $90 - 10 = 80 \text{ kJ mol}^{-1}$.

For the forward reaction, $\Delta H = \text{energy of products} - \text{energy of reactants} = 10 - 40 = -30 \text{ kJ mol}^{-1}$. The catalyst changes the pathway, not the endpoint energies. **Answer: B.**

Question 07

A sample of element X contains only the isotopes ^{54}X , ^{56}X and ^{58}X . There are twice as many ^{58}X atoms as ^{54}X atoms. The relative atomic mass of X in this sample is 56.2.

What percentage of the atoms are ^{56}X ? Treat each isotope mass as its mass number.

A 10%

B 20%

C 60%

D 70%

E 80%

Worked solution

Answer D

Let the fraction of ^{54}X be x . The fractions of ^{58}X and ^{56}X are then $2x$ and $1 - 3x$.

The weighted mean is $54x + 56(1 - 3x) + 58(2x) = 56 + 2x = 56.2$. Hence $x = 0.10$.

The fraction of ^{56}X is $1 - 3(0.10) = 0.70$, or 70%. **Answer: D.**

Question 08

Copper(II) ions react with iodide ions to form solid copper(I) iodide and iodine. Use the processes below.



A total of 0.0150 mol of Cu^{2+} reacts, and all of this copper ends in CuI. Ignore subsequent reactions of iodine. Which row is correct?

- A** I^{-} consumed: 0.0150 mol; I_2 formed: 0.00750 mol; reducing agent: I^{-} .
- B** I^{-} consumed: 0.0300 mol; I_2 formed: 0.0150 mol; reducing agent: I^{-} .
- C** I^{-} consumed: 0.0300 mol; I_2 formed: 0.00750 mol; reducing agent: I^{-} .
- D** I^{-} consumed: 0.0300 mol; I_2 formed: 0.00750 mol; reducing agent: Cu^{2+} .
- E** I^{-} consumed: 0.0450 mol; I_2 formed: 0.0150 mol; reducing agent: Cu^{2+} .

Worked solution

Answer C

Each Cu^{2+} gains one electron, so 0.0150 mol of electrons is needed. Oxidation of 0.0150 mol of I^{-} supplies these electrons and makes 0.00750 mol of I_2 .

A further 0.0150 mol of I^{-} combines with the Cu^{+} to make CuI. The total iodide consumption is therefore 0.0300 mol. Iodide loses electrons and is the reducing agent.

The overall equation is $2\text{Cu}^{2+} + 4\text{I}^{-} \rightarrow 2\text{CuI(s)} + \text{I}_2$. **Answer: C.**

Question 09

An addition polymer is made using only ethene, C_2H_4 , and chloroethene, $\text{C}_2\text{H}_3\text{Cl}$. No atoms are lost when either monomer joins the chain.

A representative section of the polymer contains 120 carbon atoms and 20 chlorine atoms. Ignore the ends of the polymer chain.

What fraction of the incorporated monomer molecules were chloroethene, and how many hydrogen atoms are in this section?

A Fraction $1/6$; 220 hydrogen atoms

B Fraction $1/3$; 240 hydrogen atoms

C Fraction $1/2$; 200 hydrogen atoms

D Fraction $2/3$; 200 hydrogen atoms

E Fraction $1/3$; 220 hydrogen atoms

Worked solution

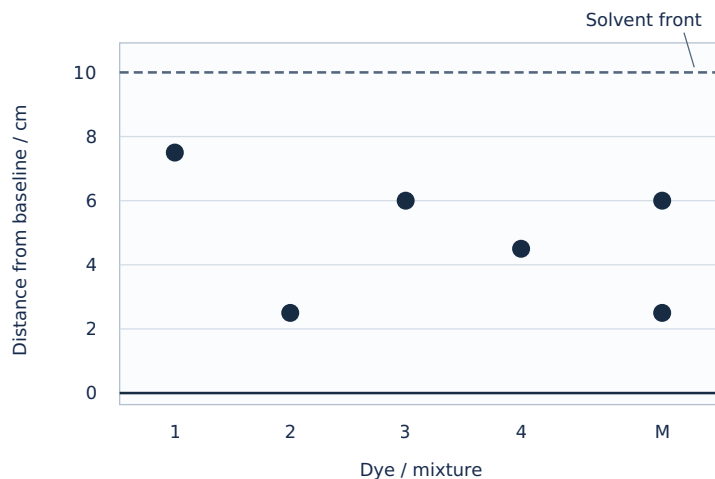
Answer E

Both monomers supply two carbon atoms, so 120 carbon atoms represent 60 monomer molecules. Only chloroethene supplies chlorine; 20 chlorine atoms therefore represent 20 chloroethene molecules.

The chloroethene fraction is $20/60 = 1/3$. The section contains 40 ethene units and 20 chloroethene units, giving $40 \times 4 + 20 \times 3 = 220$ hydrogen atoms. **Answer: E.**

Question 10

Four pure dyes and mixture M are run on the same chromatogram. M contains only one or more of these four dyes, and every component present gives a visible spot under these conditions.



Consider the statements.

1. The R_f value of dye 3 is 0.60.
2. The filter paper is the mobile phase.
3. M contains dyes 2 and 3.

Which statements are correct?

A 1 only

B 2 only

C 3 only

D 1 and 3 only

E 1, 2 and 3

Worked solution

Answer D

Dye 3 has travelled 6.0 cm and the solvent front 10.0 cm, so $R_f = 6.0/10.0 = 0.60$. Statement 1 is true.

The solvent is the mobile phase; the paper is stationary, so statement 2 is false. The spots in M match dyes 2 and 3. The restriction to these four dyes, together with the stated visibility of all components, makes this identification justified. Statement 3 is true. **Answer: D.**

Question 11

A dry mixture of helium and carbon dioxide has mass 4.00 g and occupies 2.40 dm³ at room temperature and pressure.

Treat both gases as ideal. Use a molar gas volume of 24.0 dm³ mol⁻¹, $A_r(\text{He}) = 4$ and $M_r(\text{CO}_2) = 44$.

What volume would the carbon dioxide alone occupy at the same temperature and pressure?

A 0.240 dm³

B 1.20 dm³

C 2.16 dm³

D 1.92 dm³

E 2.40 dm³

Worked solution

Answer C

The total amount of gas is $2.40/24.0 = 0.100$ mol. Let the amount of CO₂ be x mol, leaving $(0.100 - x)$ mol of helium.

The mass equation is $44x + 4(0.100 - x) = 4.00$. Thus $40x = 3.60$ and $x = 0.0900$ mol.

The carbon dioxide volume is $0.0900 \times 24.0 = 2.16$ dm³. **Answer: C.**

Question 12

A 25.0 cm^3 sample of sodium hydroxide solution is mixed with 40.0 cm^3 of $0.150 \text{ mol dm}^{-3}$ hydrochloric acid. The acid is in excess.

The remaining acid requires 10.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ sodium hydroxide for complete neutralisation.

What was the concentration of the original sodium hydroxide solution?

A $0.0800 \text{ mol dm}^{-3}$

B $0.160 \text{ mol dm}^{-3}$

C $0.240 \text{ mol dm}^{-3}$

D $0.320 \text{ mol dm}^{-3}$

E $0.400 \text{ mol dm}^{-3}$

Worked solution

Answer B

The initial amount of HCl is $0.0400 \times 0.150 = 0.00600 \text{ mol}$. The later titration uses $0.0100 \times 0.200 = 0.00200 \text{ mol}$ of NaOH, so 0.00200 mol of HCl remained.

The original sample neutralised $0.00600 - 0.00200 = 0.00400 \text{ mol}$ of HCl. Since HCl and NaOH react 1:1, it contained 0.00400 mol of NaOH.

Its concentration was $0.00400/0.0250 = 0.160 \text{ mol dm}^{-3}$. **Answer: B.**

Question 13

Chlorine reacts with hot concentrated sodium hydroxide. All the chlorine atoms end in chloride ions, Cl^- , or chlorate ions, ClO_3^- . No other element changes oxidation state.

If 0.120 mol of Cl_2 reacts, how many moles of Cl^- form?

A 0.0200 mol

B 0.0400 mol

C 0.100 mol

D 0.120 mol

E 0.200 mol

Worked solution

Answer E

Chlorine starts with oxidation state 0. It changes to -1 in Cl^- , gaining one electron per atom. In ClO_3^- , chlorine is $+5$ because three oxygen atoms contribute -6 and the ion has charge -1 .

Each chlorine atom oxidised to $+5$ loses five electrons, so five chlorine atoms must be reduced to -1 . The chloride:chlorate ratio is 5:1.

The 0.120 mol of Cl_2 contains 0.240 mol of chlorine atoms. Five-sixths enter chloride, giving $(5/6) \times 0.240 = 0.200$ mol. **Answer: E.**

Question 14

A carbonate reacts with dilute acid at constant temperature. Only carbon dioxide escapes from the flask. The cumulative decrease in the balance reading is recorded.

Time / s	0	20	40	60	80
Mass lost / mg	0	16	28	34	36

Which row correctly gives the average mass-loss rates over 0–40 s and 40–80 s, together with a possible explanation for the later change in rate?

- A** 0.70 and 0.20 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.
- B** 0.70 and 0.20 mg s⁻¹; acid depletion lowers the fraction of acid particles with energy above the activation energy.
- C** 0.70 and 0.20 mg s⁻¹; acid depletion lowers the mean kinetic energy of the remaining acid particles.
- D** 1.40 and 0.40 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.
- E** 0.20 and 0.70 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.

Worked solution

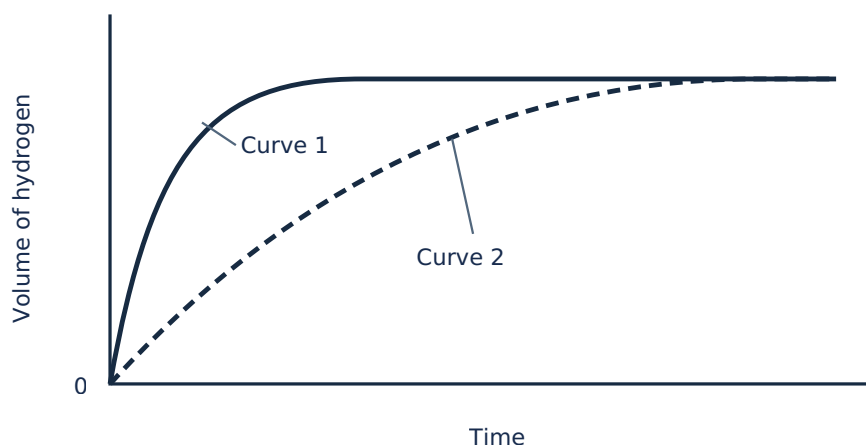
Answer A

The first average rate is $28/40 = 0.70 \text{ mg s}^{-1}$. The later interval loses an additional $36 - 28 = 8 \text{ mg}$ in 40 s, giving $8/40 = 0.20 \text{ mg s}^{-1}$.

As acid is consumed, fewer acid particles per unit volume can reach the carbonate, reducing collision frequency. At constant temperature, acid depletion does not by itself lower the mean kinetic energy or the fraction of particles above a fixed activation energy. **Answer: A.**

Question 15

Curve 2 shows hydrogen collected when magnesium chips react with excess dilute hydrochloric acid. All magnesium eventually reacts.



Each of the following changes is considered separately.

1. Use the same mass of magnesium as a powder.
2. Use twice the mass of magnesium chips of the same size.
3. Use the original magnesium chips but double the acid concentration, keeping the acid volume the same.

Acid remains in excess in every experiment. Temperature is kept constant, and all gas volumes are measured at the same temperature and pressure. Which changes could produce curve 1?

A 1 only

B 2 only

C 3 only

D 1 and 2 only

E 1 and 3 only

F 2 and 3 only

Worked solution

Answer E

Curve 1 has a higher initial rate but the same final hydrogen volume. With acid in excess, the amount of magnesium fixes the final amount of hydrogen.

Change 1 increases magnesium surface area without changing its amount, so it can increase rate while preserving the final volume. Change 3 increases acid concentration and collision frequency, again without changing the limiting amount of magnesium.

Change 2 doubles the amount of magnesium and would double the final hydrogen amount. It cannot give the shared plateau. Thus changes 1 and 3 are suitable.

Answer: E.

Question 16

The solubility of a non-volatile salt X is given below.

Temperature / °C	Solubility / g of X per 100 g of water
60	60
20	20

A 200 g sample of saturated solution at 60 °C is cooled to 20 °C. During cooling, 10.0 g of water evaporates. The mixture is allowed to reach equilibrium, and the resulting crystals contain no water.

What mass of X crystallises?

A 32.0 g

B 40.0 g

C 52.0 g

D 50.0 g

E 55.0 g

Worked solution

Answer C

At 60 °C, 160 g of saturated solution contains 100 g of water and 60 g of X. The 200 g sample therefore contains 125 g of water and 75 g of X.

After evaporation, 115 g of water remains. At 20 °C this can dissolve $(20/100) \times 115 = 23.0$ g of X.

The crystallised mass is $75.0 - 23.0 = 52.0$ g. The 200 g initial mass is the solution mass, not the water mass. **Answer: C.**

Question 17

Use the following gas-phase reactions and energy changes.



The mean bond energies are $\text{H-H} = 436$, $\text{Cl-Cl} = 242$ and $\text{C-H} = 414 \text{ kJ mol}^{-1}$. Use the same mean energy for every C-H bond in this calculation.

What value of the C-Cl bond energy follows from these data?

A 225 kJ mol⁻¹

B 335 kJ mol⁻¹

C 431 kJ mol⁻¹

D 445 kJ mol⁻¹

E 519 kJ mol⁻¹

Worked solution

Answer B

Using $\Delta H = \text{energy required to break bonds} - \text{energy released on forming bonds}$, the first reaction gives $-184 = 436 + 242 - 2D(\text{H-Cl})$. Hence $D(\text{H-Cl}) = 431 \text{ kJ mol}^{-1}$.

In the second reaction, three C-H bonds occur on both sides and cancel under the stated mean-bond approximation. The net changes break one C-H and one Cl-Cl bond and form one C-Cl and one H-Cl bond.

Therefore $-110 = 414 + 242 - D(\text{C-Cl}) - 431$, giving $D(\text{C-Cl}) = 335 \text{ kJ mol}^{-1}$.

Answer: B.

Question 18

A 100 g mixture of hexane and octane contains 40.0% hexane by mass. During fractional distillation, the first 20.0 g of distillate collected contains 75.0% hexane by mass. Assume all material is either in the flask or in the collected distillate.

Hexane boils at 69 °C and octane at 126 °C. Which row gives the percentage of hexane by mass remaining in the flask and identifies the more volatile component?

A 25.0%; hexane

B 31.25%; octane

C 50.0%; hexane

D 31.25%; hexane

E 75.0%; octane

Worked solution

Answer D

The original mixture contains 40.0 g of hexane. The distillate removes $0.750 \times 20.0 = 15.0$ g of hexane, leaving 25.0 g.

The flask now contains $100 - 20.0 = 80.0$ g of mixture, so its hexane content is $(25.0/80.0) \times 100\% = 31.25\%$.

Hexane has the lower boiling point and is the more volatile component, consistent with its enrichment in the first distillate. The stated boiling points alone would not determine the exact temperature of the mixture in the flask. **Answer: D.**

Question 19

An 80.0 g sample of anhydrous CuSO_4 , $M_r = 160$, is dissolved to make 500 cm^3 of solution. The solution is electrolysed using inert electrodes, and 8.0 g of copper is deposited at the cathode.

Assume that no copper enters the solution at the anode, that the solution volume remains 500 cm^3 , and that the only process removing Cu^{2+} is its reduction to copper metal. Use $A_r(\text{Cu}) = 64$.

What is the final concentration of Cu^{2+} ions?

A 0.25 mol dm^{-3}

B 0.50 mol dm^{-3}

C 1.00 mol dm^{-3}

D 1.25 mol dm^{-3}

E 0.75 mol dm^{-3}

Worked solution

Answer E

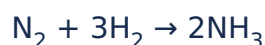
Initially, $n(\text{Cu}^{2+}) = n(\text{CuSO}_4) = 80.0/160 = 0.500 \text{ mol}$. The copper deposit is $8.0/64 = 0.125 \text{ mol}$, so 0.125 mol of Cu^{2+} has been removed.

There are $0.500 - 0.125 = 0.375 \text{ mol}$ of Cu^{2+} left in 0.500 dm^3 . Its concentration is $0.375/0.500 = 0.750 \text{ mol dm}^{-3}$.

This calculation tracks copper ions. It does not mean that sulfate ions have been removed together with the deposited copper. **Answer: E.**

Question 20

A sample of nitrogen occupies 4.80 dm^3 at room temperature and pressure. It reacts with excess hydrogen according to:



In this stage, 75.0% of the nitrogen reacts. All the ammonia produced is then converted to ammonium nitrate:



Only 80.0% of the ammonium nitrate formed is recovered as a product. Use a molar gas volume of $24.0 \text{ dm}^3 \text{ mol}^{-1}$ at room temperature and pressure and $M_r(\text{NH}_4\text{NO}_3) = 80$.

What mass is recovered?

A 9.60 g

B 14.4 g

C 19.2 g

D 24.0 g

E 25.6 g

Worked solution

Answer C

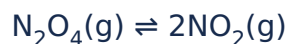
The initial amount of nitrogen is $4.80/24.0 = 0.200 \text{ mol}$. The amount that reacts is $0.750 \times 0.200 = 0.150 \text{ mol}$.

Each mole of nitrogen produces two moles of ammonia, so 0.300 mol of NH_3 forms. This makes 0.300 mol of ammonium nitrate, with mass $0.300 \times 80 = 24.0 \text{ g}$.

The recovered mass is $0.800 \times 24.0 = 19.2 \text{ g}$. The first percentage applies to nitrogen conversion and the second to recovery of the salt that actually forms. **Answer: C.**

Question 21

A sealed vessel contains an equilibrium mixture of the ideal gases N_2O_4 and NO_2 :



The vessel volume is suddenly halved at constant temperature. Consider the instant after compression, before reaction changes the amounts, and then the new equilibrium.

Which row correctly describes the NO_2 concentration at that instant, and the subsequent change in the **number of NO_2 molecules** as equilibrium is re-established?

- A** Concentration doubles; number of molecules subsequently decreases.
- B** Concentration doubles; number of molecules subsequently increases.
- C** Concentration is unchanged; number of molecules subsequently decreases.
- D** Concentration halves; number of molecules subsequently increases.
- E** Concentration doubles; number of molecules subsequently stays unchanged.

Worked solution

Answer A

Immediately after compression, the amount of NO_2 has not yet changed but the volume is half as large. Its concentration therefore doubles.

The increased pressure favours the side with fewer gas molecules. This is the left side, with one N_2O_4 for every two NO_2 on the right. The subsequent net reaction consumes NO_2 , so the number of NO_2 molecules decreases.

The question distinguishes a concentration change caused immediately by compression from a later change in chemical amount. **Answer: A.**

Question 22

A 20.0 cm^3 mixture containing only methane, CH_4 , and ethane, C_2H_6 , is burned completely in excess oxygen. It produces 30.0 cm^3 of carbon dioxide.

All gas volumes are compared at the same temperature and pressure. What volume of oxygen is consumed?

A 40.0 cm^3

B 45.0 cm^3

C 50.0 cm^3

D 55.0 cm^3

E 60.0 cm^3

Worked solution

Answer D

Let methane and ethane volumes be x and y . The total volume gives $x + y = 20.0$. Methane produces one CO_2 per molecule and ethane two, so $x + 2y = 30.0$. Thus both gases occupy 10.0 cm^3 .

The combustion equations are $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ and $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$.

The oxygen volume is $2(10.0) + (7/2)(10.0) = 55.0 \text{ cm}^3$. **Answer: D.**

Question 23

Metal P displaces Q from an aqueous salt solution, and metal Q displaces R. Carbon reduces an oxide of R but cannot reduce an oxide of Q under appropriate extraction conditions.

Use the usual reactivity-series model: an element displaces a less reactive element from its compound. What is the order from most to least reactive?

A $P > C > Q > R$

B $P > Q > C > R$

C $C > P > Q > R$

D $P > Q > R > C$

E $Q > P > C > R$

Worked solution

Answer B

The metal displacements give $P > Q > R$. Carbon reducing R oxide places C above R, while its inability to reduce Q oxide places C below Q within the stated reactivity-series model.

Combining these relations gives $P > Q > C > R$. Each neighbouring relation is needed to distinguish the alternatives. **Answer: B.**

Question 24

An impure sample contains calcium carbonate and material that does not react with acid. Excess acid produces 0.600 dm^3 of carbon dioxide at room temperature and pressure. All this gas comes from the calcium carbonate and is collected.

The sample is 75.0% calcium carbonate by mass. Use a molar gas volume of $24.0 \text{ dm}^3 \text{ mol}^{-1}$ and $M_r(\text{CaCO}_3) = 100$.

What was the sample mass, to the nearest 0.01 g?

A 2.00 g

B 2.67 g

C 3.33 g

D 4.00 g

E 5.00 g

Worked solution

Answer C

$\text{CaCO}_3 + 2\text{H}^+ \rightarrow \text{Ca}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$ gives one mole of CO_2 per mole of calcium carbonate.

The gas amount is $0.600/24.0 = 0.0250 \text{ mol}$, corresponding to $0.0250 \times 100 = 2.50 \text{ g}$ of CaCO_3 .

This is 75.0% of the sample mass, so the sample mass is $2.50/0.750 = 3.333\dots \text{ g}$, or 3.33 g. **Answer: C.**

Question 25

An aqueous solution contains one dissolved salt. Separate portions are tested.

- Sodium hydroxide gives a blue precipitate that is insoluble in excess sodium hydroxide.
- After acidification with dilute nitric acid, silver nitrate gives a cream precipitate.

Which salt is most consistent with both observations?

A CuCl

B CuCl₂

C FeBr₂

D CuBr₂

E NaBr

Worked solution

Answer D

The blue precipitate is consistent with copper(II) hydroxide, identifying Cu²⁺ ions. The cream silver-halide precipitate is silver bromide, identifying Br⁻ ions.

One Cu²⁺ requires two Br⁻ ions for charge balance, giving CuBr₂. Dilute nitric acid is specified so that the acid itself does not introduce chloride ions. **Answer: D.**

Question 26

A catalytic converter promotes the reaction:



When 0.100 mol of NO reacts, which row correctly identifies the reducing agent and the amount of electrons transferred from that agent to the species being reduced?

A CO; 0.100 mol of electrons

B NO; 0.100 mol of electrons

C NO; 0.200 mol of electrons

D CO; 0.400 mol of electrons

E CO; 0.200 mol of electrons

Worked solution

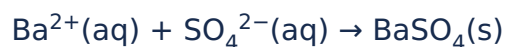
Answer E

Carbon changes from +2 in CO to +4 in CO₂. It loses two electrons per carbon atom, so CO is oxidised and acts as the reducing agent.

Nitrogen changes from +2 in NO to 0 in N₂, gaining two electrons per nitrogen atom. Therefore 0.100 mol of NO gains $2 \times 0.100 = 0.200$ mol of electrons. The same electron amount is supplied by CO. **Answer: E.**

Question 27

A 20.0 cm^3 sample of $0.100 \text{ mol dm}^{-3} \text{ BaCl}_2$ is mixed with 30.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4$. Barium sulfate precipitates:



Assume precipitation is complete and use $M_r(\text{BaSO}_4) = 233$. What mass of precipitate forms?

A 0.233 g

B 0.466 g

C 0.699 g

D 1.17 g

E 2.33 g

Worked solution

Answer B

The solutions contain $n(\text{Ba}^{2+}) = 0.0200 \times 0.100 = 0.00200 \text{ mol}$ and $n(\text{SO}_4^{2-}) = 0.0300 \times 0.100 = 0.00300 \text{ mol}$.

The precipitation ratio is 1:1. Barium ions are limiting, so 0.00200 mol of BaSO_4 forms. Its mass is $0.00200 \times 233 = 0.466 \text{ g}$. **Answer: B.**

MODULE 03

Biology

QUESTIONS & WORKED SOLUTIONS

Chemistry + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

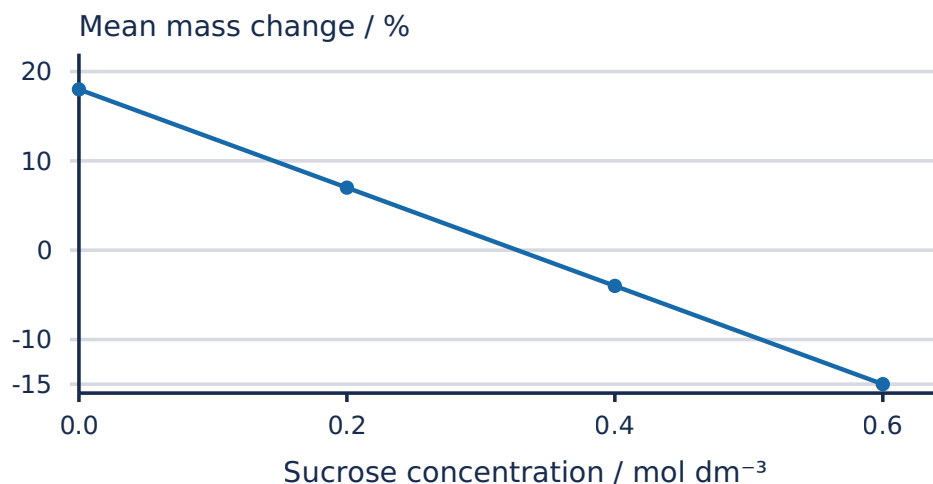
Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

Potato cylinders were placed in sucrose solutions for 2 hours. The mean percentage change in mass was measured.

Sucrose / mol dm ⁻³	0.0	0.2	0.4	0.6
Change in mass / %	+18	+7	-4	-15



Assuming the relationship is linear between the two concentrations that bracket zero change, which external sucrose concentration would give zero mean percentage change in mass?

A 0.20 mol dm⁻³

B 0.27 mol dm⁻³

C 0.33 mol dm⁻³

D 0.45 mol dm⁻³

E 0.60 mol dm⁻³

Worked solution

Answer C

Zero change is between 0.2 and 0.4 mol dm⁻³. The measured change falls by 11 percentage points over this interval; reaching zero requires a fall of 7.

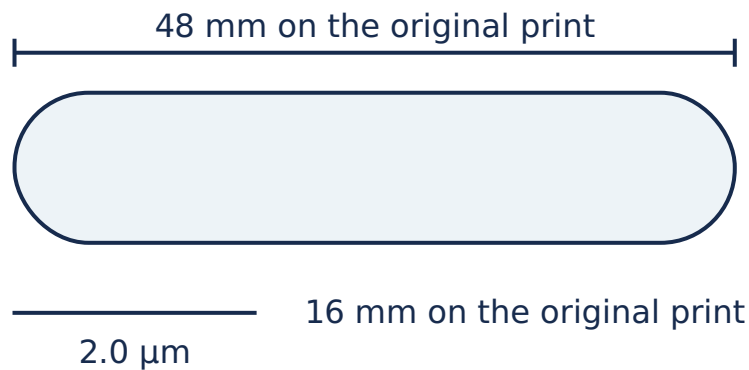
Concentration = $0.2 + (7/11) \times 0.2 = 0.327... \text{ mol dm}^{-3}$, giving **C, 0.33 mol dm⁻³**.

This estimates the external concentration giving no net mass change under the experiment's conditions.

Useful approach: Interpolate between the two measurements on opposite sides of zero.

Question 02

On an original printed micrograph, a scale bar labelled 2.0 μm measures 16 mm and a bacterium measures 48 mm. These measurements are shown schematically below; do not measure this drawing.



Which row gives the magnification of the original print, the actual bacterium length and a structure present in a typical bacterium?

A 8.0×10^2 ; 6.0 μm ; a cell wall

B 8.0×10^3 ; 6.0 μm ; a nucleus

C 8.0×10^4 ; 6.0 μm ; a nucleus

D 8.0×10^3 ; 0.6 μm ; a cell wall

E 8.0×10^3 ; 6.0 μm ; a cell wall

Worked solution

Answer E

The scale-bar length is 16 mm = 16 000 μm , so magnification = $16\,000 / 2.0 = 8.0 \times 10^3$.

The bacterium is $48/16 = 3$ times the scale-bar length, so its actual length is $3 \times 2.0 = 6.0 \mu\text{m}$. A typical bacterium has a cell wall and lacks a membrane-bound nucleus. Therefore **E** is correct.

Useful approach: Magnification uses the stated original-print measurement; actual length can be found from the scale-bar ratio.

Question 03

Nitrate uptake by root tissue was measured for 30 minutes. The external nitrate concentration was kept constant and greater than the concentration inside the cells.

Condition	Uptake / $\mu\text{mol g}^{-1} \text{h}^{-1}$
Normal oxygen, no inhibitor	18
Low oxygen, no inhibitor	8
Normal oxygen, respiratory inhibitor	5
Low oxygen, respiratory inhibitor	4

Consider the following statements.

1. Uptake decreases when oxygen availability is reduced.
2. The results support an energy-dependent component of nitrate uptake.
3. Reducing oxygen has the same absolute effect on uptake with and without the inhibitor.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer E

Uptake falls when oxygen is reduced: 18 to 8 without inhibitor, and 5 to 4 with inhibitor. Statement 1 is true. Reduced uptake with low oxygen or a respiratory inhibitor supports an energy-dependent component, so statement 2 is true.

The absolute decreases are 10 and 1 $\mu\text{mol g}^{-1} \text{h}^{-1}$, which are different. Statement 3 is false. Hence **E: 1 and 2 only**.

Useful approach: Compare the change within each inhibitor condition, rather than comparing only final values.

Question 04

A diploid organism has $2n = 6$. Count a chromosome by its centromere and count one chromatid as one DNA molecule. Immediately after DNA replication, a cell enters meiosis I. How many chromosomes and DNA molecules are present in each daughter cell immediately after meiosis I?

A 3 chromosomes and 3 DNA molecules

B 3 chromosomes and 6 DNA molecules

C 6 chromosomes and 6 DNA molecules

D 6 chromosomes and 3 DNA molecules

E 3 chromosomes and 12 DNA molecules

Worked solution

Answer B

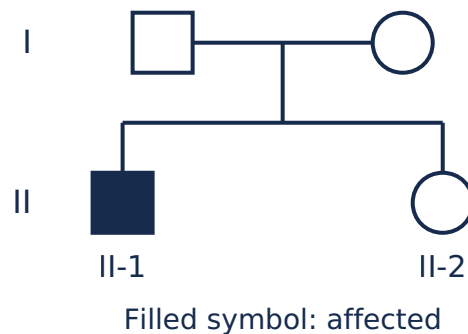
Before replication the cell has six chromosomes. Replication makes two DNA molecules (two chromatids) per chromosome, so there are 12 DNA molecules but still six chromosomes.

Meiosis I separates homologous chromosomes, not sister chromatids. Each daughter therefore receives three chromosomes, each still consisting of two chromatids: $3 \times 2 = 6$ DNA molecules. The answer is **B**.

Useful approach: Ask what is separating at each division. Meiosis I separates homologues; meiosis II separates sister chromatids.

Question 05

A rare condition is caused by an autosomal recessive allele. Two unaffected parents have an affected son and an unaffected daughter, II-2. II-2 has a child with an unrelated unaffected man. For this man, the probability of being a carrier is $1/10$. What is the probability that their child is affected?



Assume II-2 is unaffected as stated and that the father of her child is not related to her family.

A $1/10$

B $1/20$

C $1/40$

D $1/60$

E $1/120$

Worked solution

Answer D

Because the unaffected parents have an affected child, both are carriers. Their unaffected daughter has a $2/3$ probability of being a carrier: the three unaffected genotypes from a carrier $\times$ carrier cross are AA, Aa and Aa.

For an affected child, II-2 must be a carrier, the man must be a carrier, and two carrier alleles must meet:

$$(2/3) \times (1/10) \times (1/4) = 1/60.$$

Therefore **D** is correct.

Useful approach: Condition the daughter's genotype on being unaffected before multiplying by the unrelated partner's carrier risk.

Question 06

In a plant, purple flowers (P) are dominant to white flowers (p). A purple plant A is crossed with a homozygous white plant, and a large offspring sample contains equal numbers of purple and white plants. Plant A is then crossed with a known heterozygous purple plant B.

What is the probability that exactly two of the next three offspring are white?

A 9/64

B 3/16

C 1/16

D 1/4

E 27/64

Worked solution

Answer A

The first cross is $A \times pp$ and gives equal purple and white offspring, so A must be Pp. Plant B is also Pp. Each offspring is white with probability $1/4$.

Exactly two whites in three births can occur in three orders. Hence:

$$3 \times (1/4)^2 \times (3/4) = 9/64.$$

The answer is **A**; the factor of 3 is missed if one specific order is counted.

Useful approach: Count the three possible positions of the non-white offspring as well as the probability of one order.

Question 07

The coding DNA strand of a gene is shown below. The first triplet is the only part that changes in either variant.

normal: 5'-GAA CCT TGC-3'

variant 1: 5'-GAG CCT TGC-3'

variant 2: 5'-GAT CCT TGC-3'

Use the information that GAA and GAG code for glutamic acid, whereas GAT codes for aspartic acid. Which statement is correct?

- A** Both variants change the first amino acid, but neither changes the reading frame.
- B** Variant 1 leaves the first amino acid unchanged; variant 2 changes it.
- C** Variant 1 changes the first amino acid; variant 2 leaves it unchanged.
- D** Variant 1 shifts the reading frame; variant 2 changes only one amino acid.
- E** Both variants leave the entire amino-acid sequence unchanged.

Worked solution

Answer B

Variant 1 changes GAA to GAG. The supplied code says both specify glutamic acid, so it is a base substitution with no change to that amino acid (a silent substitution).

Variant 2 changes GAA to GAT, which changes glutamic acid to aspartic acid. It is therefore a missense substitution. Neither change adds or removes a base, so neither is a frameshift. **B** is correct.

Useful approach: Check whether the number of bases changes before considering a frameshift; then translate only the codon that changed.

Question 08

A gene and a plasmid are cut with a restriction endonuclease that leaves sticky ends. The diagram shows two ends that can be joined; 5' and 3' indicate strand direction.



Only the DNA near each cut end is shown.

Which sequence of events joins these ends?

- A** The overhangs pair by hydrogen bonds; a restriction endonuclease then joins the backbones.
- B** The overhangs pair by covalent bonds; DNA ligase then joins the backbones.
- C** DNA ligase first hydrogen-bonds the backbones; complementary bases then form covalent bonds.
- D** The overhangs pair by hydrogen bonds; DNA ligase then forms covalent bonds in the backbones.
- E** DNA polymerase pairs the overhangs; a restriction endonuclease then removes the paired bases.

Worked solution

Answer D

The complementary A-T base pairs form hydrogen bonds. The two strands run in opposite directions, as shown by their 5' and 3' ends.

DNA ligase seals the remaining breaks by forming covalent bonds in the sugar-phosphate backbones. The correct sequence is **D**. A restriction endonuclease cuts DNA; it does not seal the backbone.

Useful approach: Distinguish bonds between paired bases from bonds within a DNA strand.

Question 09

Matched samples of a bacterial population are tested before and after antibiotic exposure. Colony counts measure viable bacteria of each type.

	Susceptible	Resistant
Before exposure	9400	600
After exposure	50	550

Which row gives the resistant fraction among the surviving bacteria and the process that best explains its change?

- A** 5.5%; selection favouring resistant bacteria already present.
- B** About 92%; preferential survival of susceptible bacteria.
- C** About 92%; susceptible bacteria acquiring resistance because they need it.
- D** 6%; selection favouring resistant bacteria already present.
- E** About 92%; selection favouring resistant bacteria already present.

Worked solution

Answer E

There are $50 + 550 = 600$ surviving bacteria, of which 550 are resistant. The resistant fraction is $550/600 = 11/12 \approx 92\%$, compared with $600/10\,000 = 6\%$ before exposure.

Resistant variants were present initially and survived much more successfully. Selection therefore raises their frequency even though their absolute number decreases. The answer is **E**.

Useful approach: Use the total surviving population as the denominator, and distinguish frequency from absolute count.

Question 10

Equal enzyme samples are pre-treated for 10 minutes and then assayed with the same substrate concentration and temperature. The rate is measured immediately after transfer to the assay pH.

Pre-treatment pH	Assay pH	Rate / arbitrary units
6	6	14
9	9	2
9	6	13

Which comparison and interpretation fit the results?

- A** The first rate is 7 times the second; most activity lost at pH 9 is recovered at pH 6.
- B** The first rate is 7 times the second; pH 9 irreversibly denatures most enzyme molecules.
- C** The first rate is 7 times the second; the higher rate is caused by a higher assay substrate concentration.
- D** The second rate is 7 times the first; returning to pH 6 restores enzyme activity.
- E** The third rate is half the first; pre-treatment at pH 9 prevents recovery of activity.

Worked solution

Answer A

The rate ratio for the first two samples is $14/2 = 7$. After pH 9 pre-treatment, transfer to pH 6 gives a rate of 13, close to the control rate of 14.

Most activity therefore returns when pH is restored. The low rate at pH 9 does not imply irreversible denaturation of most enzyme molecules. Substrate concentration was held constant. Hence **A**.

Useful approach: Use the recovery experiment to distinguish reversible loss of activity from irreversible damage.

Question 11

Three membrane bags contain equal amounts of starch and protein, but different enzyme preparations. Each bag is placed in water at the same temperature and suitable pH.

The membrane allows maltose and amino acids through, but not starch, protein or enzymes. After 20 minutes, the surrounding water gives these results:

Preparation	Maltose detected	Amino acids detected
P	Yes	No
Q	No	Yes
R	Yes	Yes

Assume the preparations contain only amylase, protease, or both. Which row identifies P and Q and explains how products reach the surrounding water?

- A** P: protease; Q: amylase; products diffuse through the membrane.
- B** P: amylase; Q: protease; enzymes diffuse out and digest food outside the bags.
- C** P: amylase; Q: protease; soluble products diffuse through the membrane.
- D** P: amylase; Q: both enzymes; products pass through by osmosis.
- E** P: both enzymes; Q: protease; the membrane actively transports digestion products.

Worked solution

Answer C

Amylase digests starch to sugars including maltose, so P contains amylase. Protease digests protein to smaller products including amino acids, so Q contains protease. R contains both.

The permitted small soluble products diffuse from the bags into the surrounding water. Enzymes and undigested food molecules cannot cross the stated membrane. Therefore **C** is correct.

Useful approach: Separate digestion of large molecules from movement of the resulting small molecules.

Question 12

An athlete exercises from 5 to 20 minutes. Blood lactate concentration is measured alongside oxygen uptake.

Time / min	0	5	10	15	20	25	30
O ₂ uptake / L min ⁻¹	0.3	1.8	2.4	2.6	2.6	1.2	0.5
Lactate / mmol dm ⁻³	1.0	1.2	2.0	4.8	6.0	5.0	2.1

During which five-minute interval is the net increase in lactate concentration greatest, and what does the full pattern indicate?

- A** 5–10 minutes; anaerobic respiration stops as oxygen uptake rises.
- B** 15–20 minutes; the highest concentration marks the fastest lactate accumulation.
- C** 10–15 minutes; all respiration during exercise is anaerobic.
- D** 10–15 minutes; anaerobic respiration contributes while aerobic respiration continues.
- E** 20–25 minutes; lactate continues accumulating after exercise stops.

Worked solution

Answer D

The five-minute lactate changes are +0.2, +0.8, +2.8, +1.2, –1.0 and –2.9 mmol dm⁻³. The largest increase is therefore during **10–15 minutes**.

Oxygen uptake is elevated while lactate accumulates, so aerobic respiration continues and anaerobic respiration also contributes. A high final concentration need not mean the fastest rate of accumulation. The answer is **D**.

Useful approach: For equal time intervals, compare changes in concentration rather than final concentrations.

Question 13

A subject's cardiac output was measured at different heart rates.

Heart rate / beats min ⁻¹	60	80	100	120
Cardiac output / dm ³ min ⁻¹	4.8	6.0	7.0	6.6

What are the stroke volumes at heart rates of 80 and 120 beats min⁻¹, respectively?

A 7.5 cm³ and 5.5 cm³

B 75 cm³ and 55 cm³

C 75 cm³ and 82.5 cm³

D 80 cm³ and 55 cm³

E 750 cm³ and 550 cm³

Worked solution

Answer B

Stroke volume = cardiac output / heart rate.

At 80 beats min⁻¹: $6.0/80 = 0.075 \text{ dm}^3 = 75 \text{ cm}^3$.

At 120 beats min⁻¹: $6.6/120 = 0.055 \text{ dm}^3 = 55 \text{ cm}^3$. Therefore **B**. Higher heart rate does not necessarily mean greater stroke volume.

Useful approach: Divide output by the matching heart rate, then convert dm³ to cm³.

Question 14

For this question, a gas diffuses from higher to lower partial pressure. A larger exchange surface increases its rate of diffusion when other factors are unchanged.

The following partial pressures are measured in a healthy person.

Location	pO ₂ / kPa	pCO ₂ / kPa
Alveolar air	13.3	5.3
Blood arriving at alveolar capillaries	5.3	6.1

Consider the following statements.

1. Oxygen diffuses from alveoli to blood.
2. Carbon dioxide diffuses from blood to alveoli.
3. Reducing alveolar surface area would slow both transfers, all else equal.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer H

Oxygen moves down its partial-pressure gradient from 13.3 kPa in alveolar air to 5.3 kPa in arriving blood. Carbon dioxide moves from 6.1 kPa in blood to 5.3 kPa in alveolar air. A smaller exchange surface reduces diffusion rate if other factors are unchanged. Therefore all three statements are correct: **H**.

Useful approach: Compare oxygen and carbon dioxide separately; their gradients point in opposite directions.

Question 15

Compare the same healthy person after several hours without water with that person after drinking a large volume of water. Which row describes the dehydrated state **relative to the state after drinking**?

Row	ADH level	Collecting-duct water permeability	Urine volume	Urine concentration
A	Higher	Greater	Larger	Lower
B	Higher	Lower	Smaller	Higher
C	Higher	Greater	Smaller	Higher
D	Lower	Lower	Larger	Lower
E	Lower	Greater	Smaller	Higher

A Row A

B Row B

C Row C

D Row D

E Row E

Worked solution

Answer C

Dehydration lowers blood water potential, stimulating ADH release. Higher ADH increases the collecting ducts' permeability to water, so more water is reabsorbed.

Urine volume is consequently smaller and the urine is more concentrated than after drinking. All four changes agree only in **row C**.

Useful approach: Follow the sequence: blood water potential → ADH → water reabsorption → urine.

Question 16

Person P eats a carbohydrate-rich meal. Their blood glucose rises from 4.5 to 9.0 mmol dm⁻³ and remains high for several hours. Plasma insulin remains low. An injection of insulin is followed by a fall in blood glucose. Assume the target cells respond normally to insulin.

Which inference is best supported?

- A** Target cells are insensitive to insulin; extra insulin lowers glucose mainly by stimulating glycogen breakdown.
- B** Target cells are insensitive to insulin; the low measured insulin concentration excludes a secretion problem.
- C** Insulin secretion is excessive; the injection lowers glucose by further increasing secretion.
- D** Insulin secretion is insufficient; the injection lowers glucose mainly by increasing glycogen breakdown.
- E** Insulin secretion is insufficient; the injection promotes glucose uptake and storage as glycogen.

Worked solution

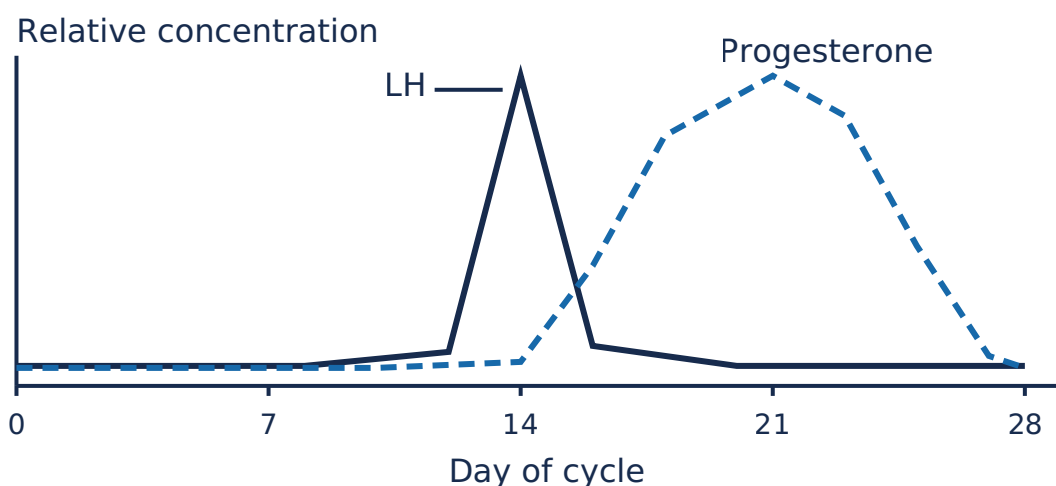
Answer E

Glucose remains high while endogenous insulin is low, but an insulin injection lowers it. With normal target-cell response, the most direct explanation is insufficient insulin secretion. Insulin promotes uptake of glucose by cells and conversion of glucose to glycogen, lowering blood concentration. **E** is correct.

Useful approach: Use both the measured hormone level and the response to the injection.

Question 17

The graph shows a typical non-pregnant menstrual cycle. Each hormone is plotted relative to its own maximum concentration.



Which statement correctly connects the hormone changes around days 14–21 with events in the ovary and uterus?

- A** The progesterone rise triggers ovulation; LH then maintains the uterine lining.
- B** The LH surge causes the uterine lining to break down; progesterone stimulates a new follicle.
- C** The LH surge releases the ovum; progesterone is then secreted by the pituitary.
- D** The LH surge triggers ovulation; progesterone from the corpus luteum helps maintain the uterine lining.
- E** The progesterone rise stimulates further LH release, allowing repeated ovulation during the same cycle.

Worked solution

Answer D

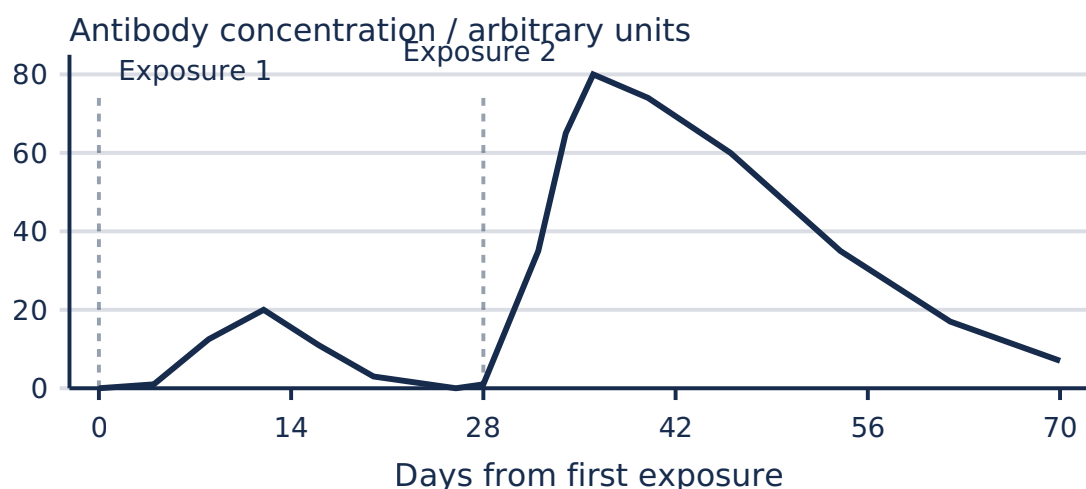
The sharp LH surge around day 14 triggers ovulation. The ruptured follicle develops into the corpus luteum, which secretes progesterone.

Progesterone helps maintain the uterine lining and contributes to negative feedback that suppresses further gonadotrophin release. Thus **D** is correct.

Useful approach: Link each hormone to its source and target rather than treating the two curves as interchangeable.

Question 18

A person receives the same dose of antigen X on days 0 and 28. The graph shows the concentration of antibodies against X.



What is the ratio of the second peak to the first, and which mechanism best explains the stronger second response?

- A** 4 : 1; antigen-specific memory lymphocytes respond on re-exposure.
- B** 2 : 1; antibodies remaining from the first response reproduce when they bind antigen.
- C** 2 : 1; antigen-specific memory lymphocytes respond on re-exposure.
- D** 4 : 1; pre-existing antibodies change their specificity after the second exposure.
- E** 4 : 1; non-specific phagocytes produce the increased antibody concentration.

Worked solution

Answer A

The peaks read from the graph are 20 and 80 arbitrary units, giving $80 : 20 = 4 : 1$.

The first exposure forms antigen-specific memory lymphocytes. Re-exposure activates a faster, larger response producing the relevant antibodies. Antibodies do not reproduce or change specificity, and phagocytes are not the cells that produce antibodies. Therefore **A**.

Useful approach: Read both peak heights from the same scale before choosing the biological explanation.

Question 19

A plant's net oxygen production was measured under four combinations of light intensity and carbon-dioxide concentration. Temperature and other conditions were kept constant.

	Low CO ₂	High CO ₂
Low light	4	5
High light	5	12

Which statement best interprets the interaction?

- A** At low light, carbon dioxide is the principal limiting factor because increasing it raises production from 4 to 5.
- B** At low carbon dioxide, raising light doubles net oxygen production.
- C** At high carbon dioxide, light is no longer limiting because increasing it changes production from 5 to 12.
- D** The effects of extra light and extra carbon dioxide are independent, because either alone changes 4 to 5.
- E** At high light, raising carbon dioxide gives a much larger increase; light limits the response to carbon dioxide at low light.

Worked solution

Answer E

At low light, raising CO₂ changes oxygen production only from 4 to 5, so light is the main limiting factor. At high light, raising CO₂ changes the rate from 5 to 12, showing that CO₂ becomes limiting once more light is available. This is **E**.

Useful approach: Compare the effect of one factor at each fixed level of the other factor.

Question 20

In a potometer, water uptake moves an air bubble through a uniform capillary tube. If the bubble moves distance h in time t , the volume uptake rate is $\pi r^2 h/t$, where r is the tube radius.

Potometer A



$$h = 8 \text{ mm}$$

$$r = 0.5 \text{ mm}$$

Solid bubble: start; dashed bubble: after 4 minutes.

Potometer B



$$h = 4 \text{ mm}$$

$$r = 1.0 \text{ mm}$$

What is the ratio of the water uptake rate in B to that in A?

A 1/4

B 1/2

C 1

D 2

E 4

Worked solution

Answer D

The measurement time is the same, so π and t cancel in the ratio:

$$B/A = (1.0^2 \times 4)/(0.5^2 \times 8) = 4/2 = \mathbf{2}, \text{ giving } \mathbf{D}.$$

Doubling the radius quadruples the cross-sectional area; halving the bubble displacement then halves this factor.

Useful approach: Bubble distance alone is not a volume: multiply by the capillary cross-sectional area.

Question 21

A complete ring of bark is removed from a tree trunk, leaving the inner wood and xylem vessels intact. Several weeks later, sugars have accumulated above the ring and the roots are losing mass.

Which explanation best accounts for both observations?

- A** Removing the bark interrupts xylem flow to the leaves; water accumulates above the ring as sugar.
- B** Removing the bark interrupts phloem transport toward the roots; xylem water transport can continue.
- C** Removing the bark interrupts phloem transport toward the leaves; the roots export their sugar supply upward.
- D** Removing the bark leaves phloem intact but prevents sugar from moving between xylem vessels.
- E** Removing the bark stops water uptake first; the intact phloem carries the roots' sugars upward.

Worked solution

Answer B

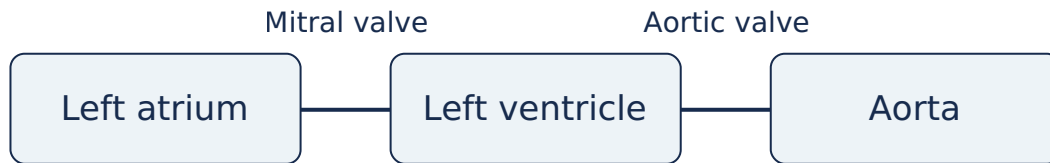
Phloem lies in the bark and transports assimilates from sources such as leaves to sinks such as roots. Removing a ring interrupts this downward translocation, causing sugar accumulation above the ring and reduced supply to roots.

The xylem is in the inner wood and has been left intact, so upward water transport can continue. This supports **B**.

Useful approach: Locate the tissue before naming the function: bark removal targets phloem, whereas the inner wood contains xylem.

Question 22

The diagram shows part of the left side of the heart. A valve opens when pressure on its upstream side exceeds pressure on its downstream side; otherwise it is closed. Ignore any delay in valve movement.



Normal permitted flow is from left to right.

Instant	Left atrium / kPa	Left ventricle / kPa	Aorta / kPa
I	1.0	0.6	10.0
II	1.0	5.0	10.0
III	1.0	13.0	12.0
IV	1.0	8.0	10.0

At which two instants are **both** valves closed?

A II and IV

B II and III

C I and II

D III and IV

E I and IV

Worked solution

Answer A

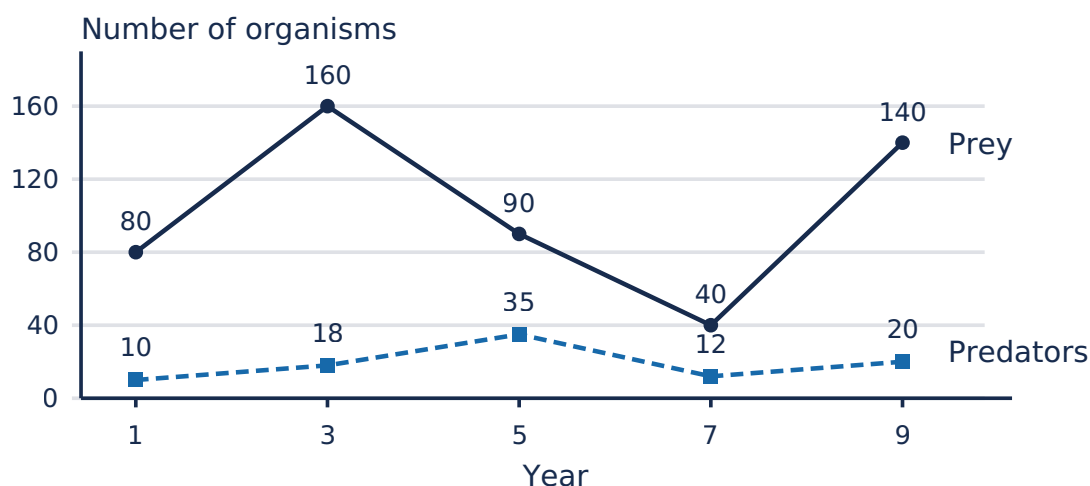
The mitral valve is closed when ventricular pressure exceeds atrial pressure. The aortic valve is closed when aortic pressure exceeds ventricular pressure.

Both conditions hold at II ($1 < 5 < 10$) and IV ($1 < 8 < 10$). At I the mitral valve is open; at III the aortic valve is open. Therefore **A: II and IV**.

Useful approach: Compare pressure across each valve separately; do not decide from ventricular pressure alone.

Question 23

Predator and prey populations are counted every two years in a closed ecosystem. Numbers beside the plotted points are the exact counts.



Consider these statements about the recorded counts.

1. The predator maximum occurs two years after the prey maximum.
2. The percentage fall in prey from year 3 to year 5 is greater than from year 5 to year 7.
3. The predator-to-prey ratio is greatest at year 5.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer F

Prey peaks at year 3 and predators at year 5, so statement 1 is true.

The two prey decreases are $70/160 = 7/16$ (less than one-half) and $50/90 = 5/9$ (more than one-half). Statement 2 is false.

Predator/prey ratios are $1/8$, $9/80$, $7/18$, $3/10$ and $1/7$. The largest is $7/18$ at year 5, so statement 3 is true. Hence **F: 1 and 3 only**.

Useful approach: A larger absolute decrease need not be a larger percentage decrease; use the starting population.

Question 24

Ten quadrats, each 0.25 m^2 , are placed at random in a meadow. The total number of plants of species A counted across all quadrats is 80. Which pair gives the estimated density and the best reason for placing the quadrats randomly?

- A** 8 plants m^{-2} ; random positions reduce investigator selection bias.
- B** 32 plants m^{-2} ; random positions reduce investigator selection bias.
- C** 32 plants m^{-2} ; random positions ensure every patch of meadow is sampled equally.
- D** 320 plants m^{-2} ; random positions reduce investigator selection bias.
- E** 20 plants m^{-2} ; random positions are chosen by starting at the densest patch.

Worked solution

Answer B

Total sampled area = $10 \times 0.25 = 2.5 \text{ m}^2$. Estimated density = $80 \div 2.5 = 32 \text{ plants m}^{-2}$. Random placement reduces the risk that the investigator chooses unusually dense or sparse patches, so **B** is correct.

Useful approach: Divide total plants by total sampled area; random placement reduces bias but cannot guarantee equal coverage.

Question 25

The table gives the energy incorporated into **new biomass each year** at the trophic levels shown for a grassland.

Trophic level	Producer	Primary consumer	Secondary consumer	Tertiary consumer
Annual production / $\text{kJ m}^{-2} \text{ year}^{-1}$	9000	1500	300	?

The energy-transfer efficiency from secondary-consumer production to tertiary-consumer production is 20%. Which row gives the missing value and a process contributing to reduced energy transfer between trophic levels?

- A** $6 \text{ kJ m}^{-2} \text{ year}^{-1}$; organic molecules are used in respiration.
- B** $60 \text{ kJ m}^{-2} \text{ year}^{-1}$; all material eaten is converted into new consumer tissue.
- C** $60 \text{ kJ m}^{-2} \text{ year}^{-1}$; some energy from respiration is transferred to the surroundings as heat.
- D** $150 \text{ kJ m}^{-2} \text{ year}^{-1}$; organic molecules are used in respiration.
- E** $1500 \text{ kJ m}^{-2} \text{ year}^{-1}$; some ingested material is egested without assimilation.

Worked solution

Answer C

Tertiary-consumer production is $0.20 \times 300 = 60 \text{ kJ m}^{-2} \text{ year}^{-1}$.

Some consumed material is not assimilated. Of the energy that is assimilated, some is released by respiration and ultimately transferred to the surroundings as heat instead of being retained in new biomass. Thus **C** gives both the value and a valid loss process.

Useful approach: Transfer efficiency compares production over the same period, rather than standing biomass measured at one instant.

Question 26

A mutation prevents a patient's liver cells from making protein P. Treatment inserts a functional gene into the patient's own stem cells and uses their descendants to replace some liver cells. A small amount of P released into the blood is sufficient. Gametes are not treated.

Which statement describes successful tissue replacement and the inheritance of this correction?

- A** The cells need to contain the gene, but need not express it; the correction is not normally inherited.
- B** The functional gene must be inserted into gametes before treated liver cells can express it; the correction is then inherited.
- C** The cells must differentiate into liver cells and express the gene; the correction is normally inherited through the patient's gametes.
- D** The cells must remain undifferentiated to replace liver cells; the correction is not normally inherited.
- E** The cells must differentiate into liver cells and express the gene; the correction is not normally inherited.

Worked solution

Answer E

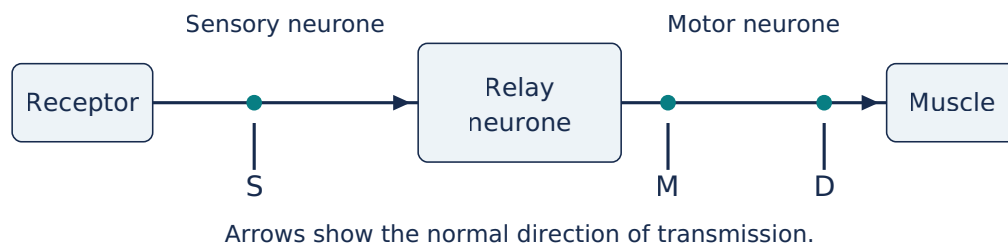
The stated aim is to replace liver cells, so the treated stem cells must produce the required differentiated descendants. Those cells must express the inserted gene to make functional protein P; carrying DNA alone is insufficient.

The correction is to somatic cells. Because the gametes are not treated, it is not normally passed to the patient's children. The answer is **E**.

Useful approach: Distinguish possession of a gene, expression of that gene and transmission through gametes.

Question 27

Electrical stimulation at S, M or D directly starts an impulse in the neurone at that point. In an intact preparation, stimulation at any of the three sites makes the muscle contract.



A single complete block to impulse transmission develops somewhere along the illustrated pathway. Stimulation does not spread to neighbouring cells. The following results are obtained:

Site stimulated	Muscle response
S	No contraction
M	No contraction
D	Contraction

Where is the block?

- A** In the motor neurone between M and D.
- B** At the connection between the relay neurone and the motor neurone.
- C** In the sensory neurone between S and the relay neurone.
- D** Between D and the muscle.
- E** Within the muscle's response to an arriving nerve impulse.

Worked solution

Answer A

Stimulation at D produces contraction, so transmission from D to the muscle and the muscle response remain functional.

Stimulation at M fails, placing the block between M and D. This also explains why stimulation at the more upstream site S fails. A block before M would be bypassed by direct stimulation at M. Therefore **A** is correct.

Useful approach: Start with the most downstream successful stimulus, then compare it with the nearest unsuccessful one.